

URL Shortener — Architecture Analysis

Project: URL Shortener

Technology: Python, FastAPI, SQLAlchemy, SQLite, Pydantic

Location: `/home/dominic/Documents/fastapiurl`

Report #: 2/20

Architecture Type

Monolithic REST API with SQLite persistence

Layers

API Layer → Service Layer → SQLAlchemy ORM → SQLite

Design Patterns

Router-Service pattern, Dependency Injection via FastAPI Depends

Scalability

Single-process, single-thread SQLite — suitable for low traffic